

Can Direct Micropayment Incentives Shape the Quality of Online Discourse?*

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October 9, 2026

Abstract

This paper studies how very small posting fees and peer-to-peer monetary rewards shape the quality and quantity of posts in an online discussion platform. We find evidence of cost-based screening, reward-driven learning, and profitability-based selection. The setting is an online discussion platform that uses Bitcoin micropayments. On the platform, users must pay a small fee to post content and can earn peer-to-peer tips from other users. Using variation in posting fees across the platform’s sub-forums and over time, we estimate how post quantity and quality change after a posting fee change. We find that the number of posts responds negatively to a posting fee increase, but that the quality of posts, as measured by tips received, responds positively to a posting fee increase. These results are consistent with a screening model in which posting costs filter out low-quality contributions. We also find evidence that users shift their posting behavior towards sub-forums where they received unexpectedly high prior feedback, demonstrating that users are responsive to tip-based feedback. Lastly, we find that users who do not recover their posting costs through tips are more likely to exit the platform. These findings provide empirical evidence that even small financial incentives can shape the quantity and quality of online discourse in a live field setting.

Keywords: micropayments; user-generated content; signaling; market design; digital platforms

JEL Classification: D82, D83, L86

*The views expressed in this paper are solely those of the authors and do not necessarily reflect the views of Stacker News. Stacker News provided the data, but did not exercise editorial control over the content of this paper, nor over the analysis, interpretation, or reporting of the results. All errors are those of the authors. The computer code used for the analysis in this paper is available at <https://github.com/ed-kung/sn-research>.

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1 Introduction

Can direct financial incentives shape the quality of online discourse? A large literature has studied how indirect mechanisms—such as ad revenue sharing, algorithmic visibility, or reputation systems—influence the quality and quantity of content that users produce (Kraut et al., 2012; Luca, 2015; Aridor et al., 2024). However, direct financial mechanisms such as requiring users to pay a fee to post or allowing peer-to-peer payments between users are rare in the field and remain largely understudied. This is especially true in the market for online discussion platforms, since each individual post in a discussion platform like Reddit or X is generally of sufficiently low value that it is not worth the transaction costs to introduce payments into these systems.

Yet, there is good reason to believe that direct payment mechanisms could meaningfully influence user behavior and content quality on discussion platforms. Signaling theory suggests that posting costs can serve as a screening device, deterring low-quality contributions while allowing users with high-quality content to signal their type by paying the cost (Joseph and Thevaranjan, 2008). Empirically, Tchernichovski et al. (2019) showed that introducing a small fee to an online rating system filtered out low information signals and increased the correlation between review ratings and quality. Other empirical studies have found that financial incentives in the form of revenue sharing can influence the quantity and quality of user-generated content (Sun and Zhu, 2013; El-Komboz et al., 2023; Kerkhof, 2024). However, none of these studies have investigated the behavioral effects of posting fees and direct peer-to-peer payments in a naturally occurring discussion platform.

One reason direct payment mechanisms have not been more widely adopted is frictions in the payments infrastructure. The value of any individual contribution

is typically too small to justify credit card transaction fees, and international platforms face additional frictions related to currency exchange and cross-border banking regulations. However, the emergence of low-cost digital payment rails, such as Bitcoin’s Lightning Network, has made it easier for platforms to integrate micropayments into their core design, opening the door to incentive structures that were previously impractical.¹

This paper provides empirical evidence on how micropayment incentives shape user behavior, using data from a Bitcoin-based discussion platform called Stacker News. Stacker News is an internet discussion forum—similar in format to Hacker News or Reddit—that integrates Bitcoin micropayments into its core design through the Lightning Network. Users must pay a fee denominated in Bitcoin, usually on the order of 1-100 satoshis (about 0.1 to 10 cents at the time the research was conducted), to post content in any of the platform’s topic-specific sub-forums, called territories.² Users can also send each other peer-to-peer Bitcoin micropayments, called “zaps”, to reward each other for content that they value. The combination of posting fees and peer-to-peer tipping creates a setting in which both the cost and reward sides of content production are mediated by real financial transactions, making Stacker News a natural field setting for studying how direct payment incentives influence user behavior on discussion platforms.

We exploit variation in posting fees across territories and over time to estimate how the quantity and quality of posts change in response to a territory fee change. Using a

¹The Lightning Network is a decentralized payments network that is built on top of Bitcoin. It enables faster and lower cost Bitcoin transactions than the base layer would allow. It works by letting users maintain peer-to-peer payment channels that allow them to “float” Bitcoin between each other indefinitely, like keeping a bar tab, until final settlement occurs on the base layer. Users can pay others with whom they have no direct channel by routing payments through other channels, usually at a small percentage fee.

²A satoshi (or sat) is 1/100,000,000th of a Bitcoin. At the time the research was conducted, 1,000 sats was worth about USD \$1.

difference-in-differences framework with territory and time fixed effects, we find that raising the posting cost in a territory reduces the number of posts with an elasticity factor of -0.265 , but increases the quality of posts—measured by tips received from other users—by an elasticity factor of 0.208 . Because territory operators can change posting fees endogenously, a central empirical concern is whether operators tend to change fees during periods when quantity is already trending down or when quality is already trending up. We conduct event study and pre-trends analysis to verify that this is not the case and, consistent with our hypothesis, the behavioral response follows the fee change rather than precedes it. The results are consistent with a screening model in which users with low-quality contributions are unwilling to pay the higher cost to post, while users with high-quality contributions are willing to pay the higher cost.

Beyond the cost-based screening mechanism, we investigate whether users learn from and respond to observed tip feedback. To do this, we study how users’ posting frequency by territory responds to prior tip feedback in that territory. We find that users are more likely to post in territories for which they have received higher than expected prior tip feedback. This effect is moderated by the user’s experience level, with more experienced users responding less sharply to observed tip feedback, consistent with prior work on social learning (Conley and Udry, 2010). Interestingly, we find that recent tip signals are not more correlated with contemporaneous posting behavior than older tip signals, which rules out contemporaneous trends like news-cycle effects as confounding explanations. The results are consistent with a learning model in which users update beliefs about their territory-specific match quality when they receive higher than expected tips from their posts in that territory.

Lastly, we investigate whether financial incentives shape the composition of active users over time. We find that users who are unable to earn enough tips to cover their

posting costs are more likely to exit the platform, while users who earn more tips than they pay in posting fees are more likely to stay. This results in a temporal selection mechanism which could further improve content quality over time.

Our findings are both surprising and unsurprising. They are unsurprising in the sense that the behavioral responses to the financial incentives observed are exactly what economic theory would predict: post quantity responds negatively to posting cost (demand curves slope downwards), post quality increases after cost increases (screening theory works), and people learn from feedback. The results are surprising in the sense that the behavioral responses are detectable and robust for even very small financial incentives, on the order of 1 to 10 cents, in a natural field setting.

Our findings have policy implications for designers of user-generated content platforms. The results suggest that content quantity and quality are mediated by financial incentives, even very small ones. Direct costs to participate can be used to screen out low quality content, while peer-to-peer tipping incentivizes the production of higher quality content. While our setting is a single platform with a highly selected user community, the underlying mechanisms we identify—cost-based screening, tip-driven learning, and profitability-based selection—are general features of micropayment systems and could plausibly operate on other platforms that adopt similar designs.

2 Incentive structures on Stacker News

From its outset, Stacker News was founded with the idea of using Bitcoin micropayments to incentivize quality content and good behavior. Its motto is “Stacker News is trying to fix online communities with economics.” This makes it a natural field setting for studying how micropayment incentives shape user behavior. There are a

number of incentive structures built into the design of Stacker News.³

Direct posting cost. First, users must pay to post. The posting cost is denominated in sats and varies by territory. A territory is a sub-forum on Stacker News, usually topic-specific.⁴ Territories are created and operated by individual users on Stacker News. To create a territory, a user chooses a name for the territory, writes a description, and pays a monthly fee to Stacker News to maintain the territory. As of October 2025, the monthly fee to operate a territory is 50,000 sats (about \$50 a month at the time the research was conducted).

Once a new territory is founded, users can post in that territory by paying a posting fee which is set by the territory operator. The territory operator earns 70% of the posting fee as revenue, and the other 30% is “taxed” and later redistributed as part of a daily rewards pool.⁵ Territory operators are therefore incentivized to choose a posting fee that optimizes between post quantity, post quality, and revenue earned. If the posting fee is too low, the territory may not earn enough revenue to cover costs, and it may also attract low quality posts. On the other hand, if the fee is too high, users may be deterred from posting in the territory. Users are incentivized to post in the territory that best fits the topic of their post so as to maximize engagement and sats earned net of posting fees.

³The following description of how Stacker News works is accurate up to October 5, 2025.

⁴As of October 2025, the most popular territories are `bitcoin`, which focuses on discussions about Bitcoin; `econ`, which focuses on news and discussion about economics; `Stacker_Sports`, which focuses on sports discussion; `AI`, focused on news and discussion about artificial intelligence; and `Politics_And_Law`, focused primarily on politics and current events. The popularity of these territories shows that Stacker News is not a Bitcoin-specific message board but covers a wide range of topics.

⁵The purpose of the tax is to discourage spam. Without the tax, territory operators would be able to post costlessly and without limit in their own territories. Revenues collected from this tax are distributed each day to active users according to an algorithm that is opaque to most users.

Peer-to-peer tipping. Second, users can tip each other directly for content that they like. On Stacker News, these tips are called “zaps”. To zap an item, the user clicks a lightning bolt icon which is featured prominently on every post and comment. When the user clicks the lightning bolt, sats are sent from the user to the creator of the item. 70% of the zap amount is sent directly to the item creator’s wallet, while 21% are sent to the territory operator and 9% are sent to the daily rewards pool.⁶ Users can set a default zap amount, but they can also customize the amount they want to zap any particular post. Henceforth, we shall use the words “zap” and “tip” interchangeably.

Like Reddit upvotes, zaps are used by the platform to surface high quality content. The default view on Stacker News ranks posts based on the amount of zaps earned by the post, modified by a time decay factor to keep the view fresh. Unlike Reddit upvotes, zaps represent real monetary value transferred from the zapper to the creator of the post. The zapping system incentivizes high quality content in two ways: not only does the system surface higher quality posts to the top of the home page, but it also lets creators of high quality content get paid directly by other users for their contributions.

While it is clear how zapping incentivizes content creators, it is not as clear what incentives users have to zap. Economists who have studied tipping in non-digital contexts have argued that tipping is not well explained by considerations of future value (Azar, 2020) and have instead focused on altruistic “warm glow” effects (Andreoni, 1990; Crumpler and Grossman, 2008). However, these are contexts where interactions are generally not repeated and where the payment includes a deterministic price along with an optional tip. On Stacker News, users have repeated interactions with each

⁶The 30% tax on zaps is a form of sybil resistance: it discourages users from creating sock puppet accounts to zap their own content.

other and there is no fixed financial reward for content creators other than tips. The incentive model may therefore be closer to that of a “pay-what-you-want” system, in which tipping is rationally used to ensure that producers do not switch to a less advantageous pricing model for consumers (Mak et al., 2015). In the case of Stacker News, the consequence of not tipping could be a reduction in content production that the user enjoys. Whether users zap due to an altruistic desire to reward content that they enjoy, or by a rational incentive to support the continued production of such content, or by a desire to influence the ranking of posts, we do observe a healthy amount of zapping on Stacker News.

Hypotheses. Based on the above discussion, we hypothesize the following about the incentive structures on Stacker News. First, we hypothesize that direct costs to post will screen out low quality content. Thus, when a territory increases its posting fee, the quantity of posts should decrease but the quality of posts that get posted should increase. Second, we hypothesize that users learn from and respond to tip feedback. Thus, users will shift their posting behavior towards territories for which they have previously received higher than expected tips. Third, we hypothesize that users who are consistently unable to cover their posting fees with tips are more likely to leave the platform. We test these hypotheses one-by-one in the following sections of the paper.

3 Data

To test these hypotheses, we downloaded a snapshot of the Stacker News database on October 5, 2025. The data covers the time period from June 11, 2021 to October 5, 2025 and contains 207,219 posts and 914,266 comments (replies), for a total of

1,121,485 items posted by 11,228 users across 117 territories. Over this time period, users zapped each other 146 million sats and paid 13 million sats in posting fees (about \$146,000 and \$13,000 respectively, at the time the research was conducted).

In addition to the Stacker News data, we downloaded daily Bitcoin price data from CoinMarketCap.com. At the time the research was conducted, the price of one Bitcoin was approximately US\$100,000, which would make 1,000 sats worth approximately \$1. This is a useful rule of thumb for converting between sats and dollars when trying to evaluate the economic significance of Stacker News micropayments.⁷

Figure 1 shows the growth in Stacker News, as measured by weekly number of items posted, alongside the weekly Bitcoin price. In aggregate, both the Bitcoin price and the activity on Stacker News grew over this time period.

4 Cost-based screening

In this section, we consider our first hypothesis: whether higher posting costs lead to fewer posts in a territory, but higher quality posts.

4.1 A model of cost-based screening

We envision a model in which a user receives “post ideas” as a Poisson process with arrival rate γ . Post ideas arrive with a quality level q , which affects how other users respond to the post, and an intrinsic motivation to post it, ϵ . The level of internal motivation may be correlated with post quality, and they are drawn from a joint probability density function $f(q, \epsilon)$. We allow different users to draw from different distributions of (q, ϵ) , but for now we focus on the decisions of a single user.

⁷In the subsequent text, whenever we convert sats to dollars, we will be using a conversion rate of 1,000 sats = \$1.

By Bayes' Rule, $f(q, \epsilon) = f_{\epsilon|q}(\epsilon|q)f_q(q)$, where $f_{\epsilon|q}(\cdot)$ is the conditional density of ϵ given q , and $f_q(\cdot)$ is the marginal density of q . We assume that $f_{\epsilon|q}(\cdot)$ is strictly log-concave and weakly satisfies the monotone likelihood ratio property in ϵ with respect to q :

Assumption 1 (Log Concavity). *For each q , $f_{\epsilon|q}(\cdot|q)$ is strictly log-concave; that is, $\ln f_{\epsilon|q}(\epsilon|q)$ is strictly concave in ϵ .*

Assumption 2 (Monotone Likelihood Ratio Property). *For any $q_2 > q_1$, the likelihood ratio of the conditional densities,*

$$\ell(\epsilon; q_1, q_2) = \frac{f_{\epsilon|q}(\epsilon|q_2)}{f_{\epsilon|q}(\epsilon|q_1)}$$

is weakly increasing in ϵ .

Assumption 1 limits the heaviness of the tail of the motivation distribution. Intuitively, it ensures that the extreme tail of the motivation distribution is not thick enough to swamp quality selection effects at very high posting costs. It is not a strong assumption, and it is satisfied by most commonly used distributional families, including the normal and logistic distributions.

Assumption 2 is a non-negative dependence condition between post quality q and intrinsic motivation ϵ . Informally, it says that higher quality ideas tend to come with (weakly) higher intrinsic desire to post: when q is higher, the distribution of ϵ shifts to the right. Independence between ϵ and q is allowed as a special case, but the assumption rules out systematic negative dependence. In our context, this means that users who receive a high quality post idea are also (weakly) more likely to feel a strong intrinsic desire to post it.

Once a post is made, zaps arrive according to a Poisson process. Both the arrival

rate of zaps and the amount of each zap are dependent on the quality of the post, q . Let $\lambda(q)$ be the arrival rate as a function of q , and let $E[z|q]$ be the expected zap amount per zap conditional on q .

Supposing the user has a discount factor of ρ , the expected present value of the Poisson stream of zaps is equal to:⁸

$$V(q) = \frac{\lambda(q)}{\rho} E[z|q] \quad (1)$$

We assume $V(q)$ is continuous and strictly increasing in q , which merely says that the expected present value of zaps is higher for higher quality posts:

Assumption 3 (Monotone Returns to Quality). $V(q) = \frac{\lambda(q)}{\rho} E[z|q]$ is continuous and strictly increasing in q .

The user will make the post if their expected present value of zaps multiplied by their intrinsic motivation is greater than the posting cost c . That is, the user will make the post if and only if:

$$V(q)e^\epsilon > c \quad (2)$$

Equivalently, we can rewrite the posting rule as a log-linear inequality. The user posts if and only if:

$$\ln V(q) + \epsilon > \kappa \quad (3)$$

where we have defined $\kappa = \ln c$. From the perspective of the user, $c > 0$ is fixed and known at the time of the post.

⁸This follows from $V(q) = \int_0^\infty e^{-\rho t} \lambda(q) E[z|q] dt$.

Under these conditions, we can demonstrate cost-based screening:

Proposition 1. *Suppose Assumptions 1-3 hold. Then an increase in c :*

- (i) strictly decreases the probability that the post idea is posted, and*
- (ii) strictly increases the expected quality of posts, conditional on posting.*

Proof. A proof is provided in the appendix. □

Proposition 1 implies that when posting cost increases, users are less likely to post, but that when they do post, the average quality of the post is higher. Although the proposition is stated from the perspective of a single user, we note that as long as the global joint distribution of (q, ϵ) holds to Assumptions 1 and 2, and if Assumption 3 also holds for all users, then Proposition 1 applies in the aggregate, even if users have heterogeneous marginal densities for q and ϵ .

Proposition 1 extends directly to a setting with multiple territories if the arrival process of post ideas in each territory is a separate Poisson process. If instead users receive a single post idea and then decide which territory j to post in based on maximization of $V_j(q_j)e^{\epsilon_j} - c_j$, then the same screening intuition carries over if quality has a strong common effect on expected returns across territories, while intrinsic motivation is relatively territory-specific. In that case, a higher posting cost tends to eliminate or redirect lower-value ideas, whereas sufficiently high-quality ideas remain worth posting.

4.2 Evidence on post quality

To test the relationship between posting fee and post quality, we run regressions of the following form:

$$\ln \text{Quality}_{ijt} = \alpha \ln \text{Cost}_{ijt} + \delta_j + \gamma_t + \epsilon_{ijt} \quad (4)$$

where i indexes a post, j indexes a territory, and t indexes a year-week.⁹ Quality_{ijt} is the measure of post quality, Cost_{ijt} is the cost (in sats) of the post, δ_j is a set of dummy variables for each territory, and γ_t is a set of dummy variables for each week. The coefficient α identifies the elasticity between posting cost and post quality. We hypothesize that $\alpha > 0$.

To measure post quality, we measure the number of sats the post earned on zaps in the first 48 hours of posting.¹⁰ Zaps represent a revealed preference measure of quality, since users must pay real money to zap. The purpose of the dummy variables is to absorb any unobserved heterogeneity in territories (e.g. if some territories are more popular and thus attract more zaps), and to absorb any global time trends in the popularity of Stacker News or the generosity of its users.

The effect of posting cost is identified from differential changes in post quality across territories when the territory posting costs change differentially. For example, if the posting cost in the **econ** territory increases while the posting cost in **bitcoin** stays the same, and subsequently the average quality of posts increases in **econ** relative to that in **bitcoin**, this would suggest that the change in posting cost likely caused the change in post quality. The coefficient α is identified off the many such changes present in the data.

⁹For example, 2024w1 is a different t than 2025w1.

¹⁰By looking at a fixed time window, we can fairly compare newer posts with older posts. Moreover, 94% of zaps are earned in the first 48 hours.

To give the reader a sense of the variation in posting costs, we plot the posting cost histories for four representative territories in Figure 2. The figure shows that there is indeed substantial variation in posting fees both across and within territories.

Before running the regression, we apply the following sample selection criteria. We do not include posts in special-purpose territories that are run by Stacker News itself.¹¹ We also do not include a number of special post types that do not follow normal posting rules.¹² Lastly, we do not include any posts made in territories by the territory operator. Since 70% of posting costs go to the territory operator, the incentives for territory operators to post in their own territories differ from that of non-operators. In the data we use for the regression, the average posting cost is 51 sats, the average amount of sats earned in the first 48 hours is 364 sats, and the average number of comments in the first 48 hours is 3.3. There is substantial variation in all three: the standard deviation of the posting cost is 241 sats, of sats earned in the first 48 hours is 6,662, and of comments in the first 48 hours is 9.6.

Table 1 shows results for regressions of specification (4) where post quality is measured as zaps in the first 48 hours. Each column of Table 1 includes a different set of controls in the regression. Column 1 includes no fixed effects, and is therefore a straight OLS regression of log zaps in the first 48 hours on log posting cost. Column 2 includes territory and week fixed effects to control for any heterogeneous effects between territories or across time. Column 3 adds in another control: the log total number of posts made in territory j in week t . This controls for a potential attention dilution effect, i.e. the possibility that zaps go up after a fee increase because number

¹¹These include territories like the **jobs** territory, in which companies (usually Bitcoin-focused companies) can make job postings, or the **AMA** territory, where influential people are invited to make “Ask Me Anything” posts.

¹²These include profile posts, which users can post and edit for free, “saloon” posts, which are daily discussion threads posted by Stacker News, and freebies—a limited number of reduced-visibility posts that users are allowed to make for free.

of posts go down, resulting in less attention congestion for the posts that do get posted. The coefficient on posting cost is still robust to the inclusion of this control, suggesting that the findings are not simply driven by an attention dilution effect. Finally, column 4 includes user fixed effects to test whether the gains in post quality are driven by within-user effects or compositional effects.

Table 1 shows that there is a statistically significant and robust positive relationship between territory posting fees and post quality as measured by zaps in the first 48 hours. In terms of the magnitude of the effect, Table 1 column 3 (our preferred specification) suggests that the elasticity between territory posting cost and expected post quality is 0.208. This roughly translates to: if a territory doubles its posting cost, then the quality of posts is expected to increase by 15.5 percent. The relationship is not explained by territory specific effects, such as a spurious correlation between posting fees and the popularity of a territory’s topic, nor is it explained by global time trends in posting fees and zap behavior on Stacker News. The best explanation for the correlation between posting cost and post quality is that users will only choose to post in a higher cost territory if they think that the post is of high enough quality to warrant the cost. This is consistent with users’ self-reported perceptions of their own behavior on the platform. One Stacker News user is quoted saying:

“If I put a lot of effort into a post, I know that the posting fee will be offset by the zaps. And even if it doesn’t, I’m just happy to show the work I’ve done and I’m ready to pay for it. On the other hand, when I share a simple link with a few quick comments and quotes, it takes me a few minutes, and I look for the cheaper territory to post it. If I don’t find any cheap territory, I’d rather not post, as I don’t think paying to post

links is worth the same as paying to post for [my] actual proof-of-work.”¹³

Finally, we comment on column 4 of Table 1, where we include user fixed effects. The inclusion of user fixed effects reduces the estimated elasticity between posting cost and post quality from 0.208 to 0.100 (still statistically significant at the 5% level). This suggests that the increase in post quality in response to an increase in posting cost is driven by both intensive margin (within-user) and extensive margin (compositional) effects. On the intensive margin, the same users now make higher quality posts, conditional on posting. On the extensive margin, the composition of posters in the territory shifts towards higher quality, most likely due to lower quality users dropping out. Column 3 remains our preferred specification over column 4, because we view the effect on the composition of users as a dimension we want to capture when studying the relationship between posting cost and post quality.

4.2.1 Event study analysis

Our empirical setting is similar to a difference-in-differences (DID) regression, because we are identifying the effect of territory fee changes from the change in post quality following the fee change, while allowing for baseline differences between territories and common time trends in post quality. In classical DID, an important assumption is that the average post quality in a territory does not systematically trend upwards prior to a post fee change. If it did so, that would call into question whether the timing of the fee change is endogenous to some prior effect that was already increasing quality. Our setting is a little more complicated than a classical DID because we have multiple fee changes in a territory, bidirectional fee changes, and continuous variation in the size

¹³In this statement, the user is referring to link posts, which are posts that include a link to an external URL, often without any text in the post body. Link posts usually receive less zaps and are considered to be of lower quality because the content is not produced by the user themselves, and are thus less rewarded by zaps.

of the fee change. Nevertheless, it would be useful for us to test whether there are any systematic post quality changes prior to a fee change.

To test this, we run the following regression:

$$\begin{aligned} \ln \text{Quality}_{ijt} = & \sum_{s=-12}^{12} \alpha_s^+ 1[\Delta \text{Cost}_{j,t-s} > 0] \\ & + \sum_{s=-12}^{12} \alpha_s^- 1[\Delta \text{Cost}_{j,t-s} < 0] + \delta_j + \gamma_t + \epsilon_{ijt} \end{aligned} \quad (5)$$

Here ΔCost_{jt} is a continuous variable for how much the posting cost in territory j changed in week t , and $1[\Delta \text{Cost}_{jt} \gtrless 0]$ is an indicator for whether the posting cost in territory j increased/decreased in week t . Thus, $\alpha_s^{+/-}$ measures the contribution to post quality of being s weeks away from a posting cost increase/decrease when $s < 0$, and the contribution to post quality of being s weeks after a posting cost increase/decrease when $s > 0$.

We estimate equation (5) and plot the estimates for $\alpha_s^{+/-}$, along with their 95% confidence intervals, in Figure 3. The left panels show the results for a specification in which user fixed effects were not included, and the right panels show the results for a specification in which user fixed effects were included.

Figure 3 shows that there was no systematic upward trend in post quality prior to post fee increases. After a post fee increase, we see that post quality begins to trend upwards. By comparing the top left and top right panels, we also see that including user fixed effects moderates the effect, but does not eliminate it—suggesting the presence of both compositional and intensive margin effects. Although noisier, we find a similar result surrounding decreases in post fees. There is no systematic downward trend in post quality prior to post fee decreases, and following a post fee decrease post quality begins to trend slightly downwards.

4.3 Evidence on post quantity

Now we consider whether territory posting cost also influences the number of posts a territory receives. If users are sensitive to posting cost, then we expect that territory posting fee will be negatively correlated with number of posts. To test this, we run regressions of the following form:

$$\ln \text{Posts}_{jt} = \alpha \ln \text{Cost}_{jt} + \delta_j + \gamma_t + \epsilon_{jt} \quad (6)$$

where j indexes a territory and t indexes a week. Posts_{jt} is the number of posts posted in territory j in week t , Cost_{jt} is territory j 's posting fee in week t (measured as the average across days), δ_j is a set of territory fixed effects, and γ_t is a set of week fixed effects. The results of this regression, for various combinations of fixed effects, are presented in Table 2. As expected, there is a robust and statistically significant negative relationship between territory posting cost and the quantity of posts in a territory. Using column 3 as our preferred specification, the results imply an elasticity of post quantity with respect to post cost of -0.265 . This translates to: if the posting cost doubles, the territory can expect to receive 16.8 percent fewer posts per week.

4.3.1 Event study analysis

We now turn to an event study analysis to test whether there are systematic changes in post quantity prior to territory fee changes. We run the following regression:

$$\begin{aligned} \ln \text{Posts}_{jt} = & \sum_{s=-12}^{12} \alpha_s^+ 1[\Delta \text{Cost}_{j,t-s} > 0] \\ & + \sum_{s=-12}^{12} \alpha_s^- 1[\Delta \text{Cost}_{j,t-s} < 0] + \delta_j + \gamma_t + \epsilon_{jt} \end{aligned} \quad (7)$$

The notation and logic is much the same as in (5), so we will not repeat ourselves here. Figure 4 shows the estimated values for $\alpha_s^{+/-}$ and their 95% confidence intervals.

Figure 4 shows that there is no systematic decrease in post quantity prior to a territory fee increase. If anything, there appears to be a systematic upward trend in post quantity prior to fee increases, followed by a downward trend in post quantity after the fee increase. For negative fee changes, there appears to be a slight systematic downward trend in post quantity prior to a fee decrease, followed by an upward trend in post quantity after a fee decrease.

The fact that positive (negative) fee changes tend to follow systematic upward (downward) trends in post quantity suggests that territory operators may be endogenously adjusting their fees in response to territory patterns. For example, an operator may hike the fee if the territory is getting flooded with low quality posts. The endogeneity is not necessarily a concern for identification if the reversal in trends is *caused by* the fee change. However, it does raise the potential concern that the results are driven by mean reversion. For example, if territory activity tends to mean revert, and if territory operators tend to raise fees in response to an influx of posts and lower fees in response to a dearth of posts, then the outcomes subsequent to a fee change may be driven by mean reversion rather than the change in fees. It seems unlikely to us that the timing of mean reversion would coincide so closely with the timing of the fee change, but we nevertheless try to control for the possibility of mean reversion. In the right two panels of Figure 4, we show the event study results while controlling for lagged territory growth in equation (7).¹⁴ We still see the same pattern of a change in post quantity subsequent a fee change, even after controlling for lagged growth. We conclude that the results therefore are driven by the specific timing of fee changes

¹⁴Specifically, we control for the percent growth in territory posts between 1 week ago and 12 weeks ago.

and not by a spurious correlation between fee changes and mean reversion.

Comparing Figures 3 and 4 reveals a pattern that is consistent with our hypothesized mechanism. Prior to a fee increase, post quantity trends upwards while post quality remains flat. The absence of a pre-trend in quality indicates that quality improvements observed after a fee increase are not the continuation of a pre-trend that was already underway. Rather, quality begins to rise only after the fee change takes effect, consistent with a cost-based screening mechanism. The rising quantity prior to a fee increase suggests that territory operators may be raising fees in response to growing volume, with the quality improvement emerging as a consequence of the higher fees screening out lower quality posts. The reverse pattern surrounding fee decreases tells a symmetric story. This story appears consistent with how some territory operators on the platform have described their rationale for fee changes. One territory operator is quoted saying:

“For **science**, I only experimented very little at the beginning. Don’t remember the specifics, but I must have tried really low (like 10-20 sats) and then a bit higher (like 100 sats). Because I had a personal policy of upvoting each post that had a modicum of quality (e.g. not just a link without context), I ended up with a small army of posters (assmilkers) that were doing the bare minimum but not enough to instigate comments or discussions. I think the reason for the 100 sats was to see how this increase would affect the willingness of those low-effort posters to post. It worked, but a side effect is that I was getting much less volume, also from quality posters. So I put it back down, and just let the community decide if a post is worth getting sats.”

This operator’s reasoning reflects awareness of the cost-based screening mechanism

as a tool for managing both post quantity and post quality.

4.4 Discussion

Taken together, the results are both surprising and unsurprising. They are unsurprising in the sense that they conform with economic theory: demand curves slope downwards (when posting costs go up, number of posts goes down), and screening theory works (when posting costs go up, higher quality posts are made). They are surprising in the sense that even such small micro-incentives (the average posting cost is just 51 sats, or about 5 cents) are enough to influence user behavior in such a way that post quality is improved. The results suggest that even very small posting fees can meaningfully affect the quantity-quality tradeoff in online discussion platforms.

5 Tip-based learning

We now consider our second hypothesis: that users learn from and respond to tip feedback by shifting their posting behavior towards better rewarded post types.

5.1 Model of tip-based learning

We start by developing a model of learning from tip feedback. Because learning involves many unobservables, like users' priors, information sets, and sources of uncertainty, we aim to keep the model parsimonious. The purpose of the model is to illustrate how assumptions about the decision structure lead to testable predictions, rather than to deeply micro-found the assumptions.¹⁵

¹⁵In the appendix, we develop a more microfounded model and discuss how they map onto our assumptions here.

We assume that users are learning about their expected rewards in territories. Let i index a user and let j index a territory. The expected present value of the Poisson stream of zaps that user i receives in territory j is:

$$V(q; \Theta_{ij}) = \Theta_{ij} v(q) \quad (8)$$

where $v(q)$ is known, continuous, and strictly increasing in q . Θ_{ij} is an unknown, user-territory specific match parameter which reflects how well user i 's posts are suited for territory j , holding objective post quality fixed.

Let $\mathcal{I}_{it}$ denote user i 's information at time t , which includes previously observed tip rewards. Define A_{ijt} and α_{ijt} as the expected value and the log expected value of Θ_{ij} conditional on the user's information set:

$$A_{ijt} = E[\Theta_{ij} \mid \mathcal{I}_{it}], \quad \alpha_{ijt} = \ln A_{ijt} \quad (9)$$

Post ideas for territory j arrive for user i at rate γ_{ij} , and their quality and intrinsic motivation are distributed according to $f^{ij}(q, \epsilon) = f_{\epsilon|q}^{ij}(\epsilon|q) f_q^{ij}(q)$. As in Section 4.1, we assume that within each territory, $f^{ij}(q, \epsilon)$ satisfies log concavity and the monotone likelihood ratio property.

If a post idea for territory j arrives for user i at time t with quality and intrinsic motivation (q, ϵ) , the posting rule is post if:

$$\alpha_{ijt} + \ln v(q) + \epsilon > \kappa_{jt} \quad (10)$$

where $\kappa_{jt} = \ln c_{jt}$ is the log posting cost in territory j at time t . The probability (over

q, ϵ) that the idea is posted is given by:

$$P_{ijt} = \int S_{\epsilon|q}^{ij} (\kappa_{jt} - \alpha_{ijt} - \ln v(q) | q) f_q^{ij}(q) dq \quad (11)$$

which is decreasing in κ_{jt} and increasing in α_{ijt} . Thus, higher posting costs deter posting, but a greater expectation of tip rewards incentivizes posting. The instantaneous rate of user i posting in territory j at time t is $\gamma_{ij} P_{ijt}$.

5.2 Empirical specification

To bring the model to the data, we model belief updating at a weekly frequency. This need not imply that users formally update their beliefs once a week, but it captures the idea that behavioral responses to tip feedback are gradual and that high-frequency variation in individual posting decisions is largely idiosyncratic. The temporal aggregation also permits territory-by-week fixed effects that absorb common changes in territory-level attention and posting incentives, allowing learning to be identified from variation in individual post histories.

We also model users as updating their beliefs using tips that a post receives after 48 hours, Z . This keeps the model parsimonious, because $\ln Z$ is a noisy signal of $\theta_{ij} + \ln v(q)$, and it allows us to abstract away from details about the timing of when posts receive their zaps.¹⁶ We therefore assume that users update their beliefs according to their history of “quality-adjusted zap signals”: $r = \ln Z - \ln v(q)$.

Let $\mathcal{I}_{ijw}$ be user i ’s information set for territory j at the start of week w . Let p index a post, sorted by timestamp, and let $i(p)$ be the user who made the post, $j(p)$ be the territory of the post, and $\tau(p)$ be the timestamp of the post. Let $\tau(w)$ be the

¹⁶To see this, we note that $E[Z] \propto V(q) = \Theta_{ij} v(q)$, according to equations (1) and (8). We can therefore write $Z = \Theta_{ij} v(q) e^\eta$, or $\ln Z = \theta_{ij} + \ln v(q) + \eta$, where η is an i.i.d. residual arising from the idiosyncratic realizations of the Poisson zaps process.

timestamp of the start of a week. The information set available to user i for territory j at the start of week w is:

$$\mathcal{I}_{ijw} = \{(Z_p, q_p) : p \in \mathcal{P}_{ijw}\} \quad (12)$$

$$\mathcal{P}_{ijw} = \{p : \tau(p) < \tau(w) - 48 \text{ hours}, i(p) = i, j(p) = j\} \quad (13)$$

In words, the user is able to observe signals for all their posts made in the territory 48 hours before the start of the week.

We postulate the following reduced form specification for α_{ijt} , the log of the user's posterior mean over Θ_{ij} at the start of week w :

$$\begin{aligned} \alpha_{ijw} &= b_{ij}^0 + \beta \sum_{p \in \mathcal{P}_{ijw}} K_{ijwp} [\ln Z_p - \ln v(q_p)] \\ &= b_{ij}^0 + \beta \sum_{p \in \mathcal{P}_{ijw}} K_{ijwp} r_p \end{aligned} \quad (14)$$

In words, the log of the user's posterior mean over Θ_{ij} is a weighted sum of previously observed quality-adjusted zap signals, $r_p : p \in \mathcal{P}_{ijw}$. This specification nests standard models of learning, such as conjugate-normal Bayesian updating, but it is flexible enough to allow for effects like attention decay or drift. For example, suppose we assume that users pay more attention to recent signals. Then, we could specify the kernel weights as:

$$K_{ijwp} \propto e^{-(\tau(w) - \tau(p))/\sigma} \quad (15)$$

However, we need not specify the weights in this manner, and in practice we can try different assumptions about the information environment.

Now, let $E[N_{ijw} \mid \mathcal{I}_{ijw}]$ be the expected number of posts user i will make in ter-

ritory j in week t , conditional in the information set $\mathcal{I}_{ijw}$. Under our beliefs model, the exact form is equal to:

$$E[N_{ijw} | \mathcal{I}_{ijw}] = \gamma_{ij} \int_{\tau(w)}^{\tau(w+1)} \int S_{\epsilon|q}^{ij}(\kappa_{jt} - \alpha_{ijw} - \ln v(q) | q) f_q^{ij}(q) dq dt \quad (16)$$

This expression shows that the number of posts in week w is expected to depend on:

- User-by-territory effects; namely, a user-territory-specific arrival rate of post ideas, γ_{ij} , and user-territory-specific distribution of post quality and intrinsic motivation, $f^{ij}(q, \epsilon)$. This captures permanent differences in users' preferences and ability to post in different territories.
- Territory-by-time effects; namely, the posting cost of the territory at any time t , κ_{jt} . In the reduced form, territory-by-time effects could also capture topic-specific trends, such as those driven by news cycles, that are not user-specific.
- A weighted sum of previously observed quality-adjusted zap signals, α_{ijw} . Higher past signals will lead to a higher expected number of posts in the week.

To turn this into a reduced form estimating equation, we model the number of posts as a Poisson distribution:

$$\begin{aligned} N_{ijw} &\sim \text{Poisson}(\mu_{ijw}) \\ \ln \mu_{ijw} &= \beta X_{ijw} + \xi_{ij} + \zeta_{jw} + \nu_{iw} + \varepsilon_{ijw} \\ X_{ijw} &= \sum_{p \in \mathcal{P}_{ijw}} K_{ijwp} r_p \end{aligned} \quad (17)$$

User-by-territory fixed effects, ξ_{ij} , capture user-specific preferences and tendencies to post in specific territories. Territory-by-week fixed effects, ζ_{jw} , capture territory-specific time trends as well as variation in posting costs. We also add user-by-week

fixed effects ν_{iw} to capture changing non-territory-specific user motivation over time. The model is estimated using Poisson pseudo-maximum likelihood, as implemented in R’s `fixest::fepois` package.

A few comments on identification. First, identification of learning comes from user-level variation in territory-specific signals. If all users had access to the same signals and formed the same expectations about territory rewards, then this would be fully absorbed into ζ_{jw} . Second, the main threat to identification comes from user-by-territory specific time trends. For example, if users have time-varying territory-specific ability, then past observed signals may reflect growing ability, which is also correlated with subsequent posting. For the purposes of this paper, we do not really care whether the learning is about a fixed or temporal user-territory match; in either case the result is that tip feedback affects user posting decisions. So if users are learning from tip feedback about temporal ability, then we do not consider that a threat to identification. A correlation between past signals and subsequent posting is only spurious if the subsequent posting would have happened regardless of what the prior tip feedback was, in a way that the correlation only exists as coincidence. An example of this would be a news cycle that generates high rewards for specific topics, but that only some users are heavily interested in posting about. To the extent that the effect of the news cycle is not captured in $\ln(q)$, it would show up in both higher quality-adjusted zap signals and increased number of posts, but the increased number of posts would have happened regardless of the higher signals. Without purely exogenous variation in signals, we cannot rule out such threats to identification. Our results should therefore be interpreted as evidence consistent with learning rather than a tightly identified causal estimate of learning.

5.3 Results

We now consider estimation of equation (17). We estimate the equation with three different sets of weights:

$$X_{ijw}^U = \sum_{p \in \mathcal{P}_{ijw}} K_{ijwp}^U r_p, \quad K_{ijwp}^U \propto 1 \quad (18)$$

$$X_{ijw}^R = \sum_{p \in \mathcal{P}_{ijw}} K_{ijwp}^R r_p, \quad K_{ijwp}^R \propto e^{-(\tau(w) - \tau(p))/\sigma} \quad (19)$$

$$X_{ijw}^E = \sum_{p \in \mathcal{P}_{ijw}} K_{ijwp}^E r_p, \quad K_{ijwp}^E \propto \frac{1}{\sqrt{n_{ij}(p)}} \quad (20)$$

X_{ijw}^U is the weighted average of all user i 's past signals in territory j , with uniform weights. X_{ijw}^R puts more weight on recent signals, with an exponential time decay with half-life of 8 weeks. Users may weigh recent signals more if there is temporal drift in the learning target or if their memory decays over time. X_{ijw}^E puts more weight on early signals. $n_{ij}(p)$ is the rank order of post p in user i 's territory j post history; i.e. $n_{ij}(p) = 1$ if p is the first post, $n_{ij}(p) = 2$ if it's the second post, and so on. Earlier signals may carry more weight if attention is costly, leading to an entrenchment of beliefs after a few early signals, or if users learn the stable component of the match early and later signals mostly reflect transitory shocks.

To calculate r_p , we first estimate $\ln v(q)$ as a non-parametric function of objective post characteristics X_p . These characteristics include the word count, whether the post is a link post, the number of links and images in the post, and a 10-dimensional semantic embedding vector of the post text. The estimator minimizes the mean squared error in the equation:

$$\ln Z_p = \mu_{i(p)j(p)} + \underbrace{\delta_{w(p)} + g(X_p)}_{\ln v(q_p)} + \varepsilon_p \quad (21)$$

and we construct:

$$\begin{aligned}\hat{r}_p &= \ln Z_p - \widehat{\ln v(q)} \\ &= \ln Z_p - \hat{\delta}_{w(p)} - \hat{g}(X_p)\end{aligned}\tag{22}$$

User-territory fixed effects $\mu_{i(p)j(p)}$ are included in the regression but not included in the construction of $\hat{r}_p$ because that is the parameter users are learning about. Week fixed effects $\delta_{w(p)}$ are included in the regression to capture global time trends that are assumed to be known to users. We describe the non-parametric estimation of $\hat{g}(X_p)$ in more detail in the appendix.

The results of regression (17) using X_{ijw}^U , X_{ijw}^R , X_{ijw}^E are reported in columns 1-3 of Table 3. All three coefficients are positive and statistically significant, which means prior tip signals are positively correlated current propensity to post. To check whether recent or early signals matter more, we also estimate a model in which all three of X_{ijw}^U , X_{ijw}^R , and X_{ijw}^E are included in the regression. Since each is a weighted average of prior signals, the sum is a linear combination of the kernel weights, and we let the data tell us which weighting scheme better explains the data. The results from this combined regression are reported in column 4 of Table 3. Interestingly, the results show that the data is best fit by placing more weight on the uniform kernel than either the recency-based or the early-signals kernel. Although not statistically significant at the 10% level, the coefficients on X_{ijw}^R and X_{ijw}^E in column 4 are negative, suggesting (if anything) that users pay more attention medium-term signals than either recent signals or early signals. In column 5 of Table 3, we also test whether the effect of past signals is moderated by the user’s experience level. We measure experience as log one plus the number of posts the user has made in the territory prior to week w . We find that experienced users are somewhat less responsive to past feedback

than inexperienced users. This suggests that experienced users may have already developed sharp beliefs or habits of posting behavior, making them less responsive overall to new signals.

Using column 1 as our preferred specification (since estimates from column 4 are noisier), we estimate that a 10 percent increase in prior quality-adjusted zap signals would increase posting frequency by 0.86 percent. This is quite a modest increase, but we do note that the standard deviation of X_{ijw}^U is quite large at 1.69. A one standard deviation increase in X_{ijw}^U is estimated by the model to increase post frequency by 15.5 percent, a much more significant number.

Identification. The lack of statistical significance and the small coefficient magnitude of the recency-weighted kernel appears to rule out news-cycle and other contemporaneous effects as alternative explanations for why prior signals are correlated with current posting. If the correlation was driven by news cycles or temporal trends, we would expect recent signals to be more correlated with current posting than past signals. Instead, we find that the correlation between prior signals and current posting is best explained by a uniform average of prior signals. The main threat to identification would be an unobserved user-territory specific condition which is both (i) varying over time and (ii) causes past signals to be correlated with current posting in such a way that doesn't give more weight to either recent or early signals. Although we cannot rule out some possible confounder that satisfies these conditions, the range of alternative explanations that could match these patterns seems narrow.

5.4 Discussion

In an online message board, there are many potential unknown parameters that users are learning about, and there are many potential sources of information to learn from. We have demonstrated evidence which is consistent with one particular unknown parameter and one particular source of information: users learning about their user-territory specific match quality using quality-adjusted tip signals from their own posts. This was demonstrated by showing that users who received higher than expected feedback from their posts in a territory are more likely to subsequently post in that territory. The correlation cannot be driven by a time-invariant user-specific match quality, nor can it be driven by a contemporaneous confounder which drives both recent feedback and current posting. The most plausible explanation is that users learn continually from tip feedback and adjust their posting behavior as a result.

Of course, there are many other potential learning-driven behavioral dynamics that we did not investigate. For example, we did not investigate social learning (users learning from other users' posts), nor did we investigate learning about post characteristics (users learning about what attributes of a post are better rewarded). We think these and other dynamics are worth investigating, but we leave it to future research to do so.

6 Profitability-based selection

We now test our last hypothesis: that consistently unprofitable users are more likely to exit the platform, whereas profitable users are more likely to stay. This would arise straightforwardly out of a learning model in which some users learn that their returns are consistently low.

To measure profitability, we only consider whether the user is able to earn enough

tips and rewards to cover their posting costs. We do not consider their own tips to other users as part of their costs, since tipping is entirely optional. Moreover, some users may have altruistic motives to tip, as discussed in the introduction, so greater tipping could indicate a greater commitment to the platform.

We define a user as inactive if they go four or more consecutive weeks without making any posts, any comments, or zapping any items. The first week of inactivity is considered the week they went inactive. Inactive users can become active again, though we treat this event as exogenous.

The goal of this section is to estimate how user profitability affects the probability of becoming inactive. We estimate linear probability models of the following form:

$$\text{Inactive}_{it} = \alpha \text{Unprofitable}_{it} + \beta \text{Unprofitable}_{it} \times \text{Growth}_t + \gamma_t + \epsilon_{it} \quad (23)$$

where i indexes a user and t indexes a week. Inactive_{it} is a binary indicator for whether user i became inactive on week t , Unprofitable_{it} is a binary indicator for whether the user was unprofitable in the last 8 weeks, and Growth_t is the percentage growth in Bitcoin-USD price in the last 8 weeks. γ_t is a week fixed effect to capture global time trends, and ϵ_{it} is the error term. We hypothesize that $\alpha > 0$, indicating that users who have been unprofitable are more likely to become inactive, and that $\beta > 0$, that this effect is amplified when Bitcoin’s price has recently appreciated.

Table 4 reports the results for this regression. Column 1 shows the result when only profitability in the last 8 weeks is included as a regressor. Column 2 additionally controls for the number of items the user posted in the last 8 weeks. Unsurprisingly, users who have posted more items in the last 8 weeks are less likely to become inactive, as it shows a greater commitment to the platform. Including the number of items in the last 8 weeks significantly moderates the effect of unprofitability, suggesting that

there are some unprofitable users who post a lot anyway. Column 3 adds week fixed effects to control for global time trends, but the coefficient of interest is not much changed. Lastly, column 4 interacts unprofitability with the Bitcoin price appreciation over the last 8 weeks.

Using column 4 as the preferred specification, the results show that users who were unprofitable in the last 8 weeks are 5.5 percentage points more likely to become inactive than users who were profitable in the last 8 weeks. This is in comparison to a baseline inactive probability of 14.1% per week. In addition, column 4 shows that unprofitable users are even more likely to become inactive when the Bitcoin price has recently appreciated. If the Bitcoin price has appreciated by 10 percent in the last 8 weeks, this increases the effect of unprofitability from a 5.5 percentage-point to a 6.1 percentage-point increased probability of exit. This suggests that users are sensitive to not just their profitability as measured in sats, but also their profitability as measured in purchasing power.

7 Conclusion

Our analysis of Stacker News data confirms three central hypotheses: first, that posting costs can increase the quality of posted content by screening out lower-quality content, consistent with signaling theory; second, that early tip-based feedback shapes users' subsequent posting behavior; and third, that unprofitable users are more likely to leave the platform while profitable users are more likely to stay. This third point suggests that content quality may further increase over time as users who consistently produce low-value content eventually exit the platform.

More broadly, the results suggest that users of digital platforms are responsive to financial micro-incentives, even ones of very small monetary value. Integrating

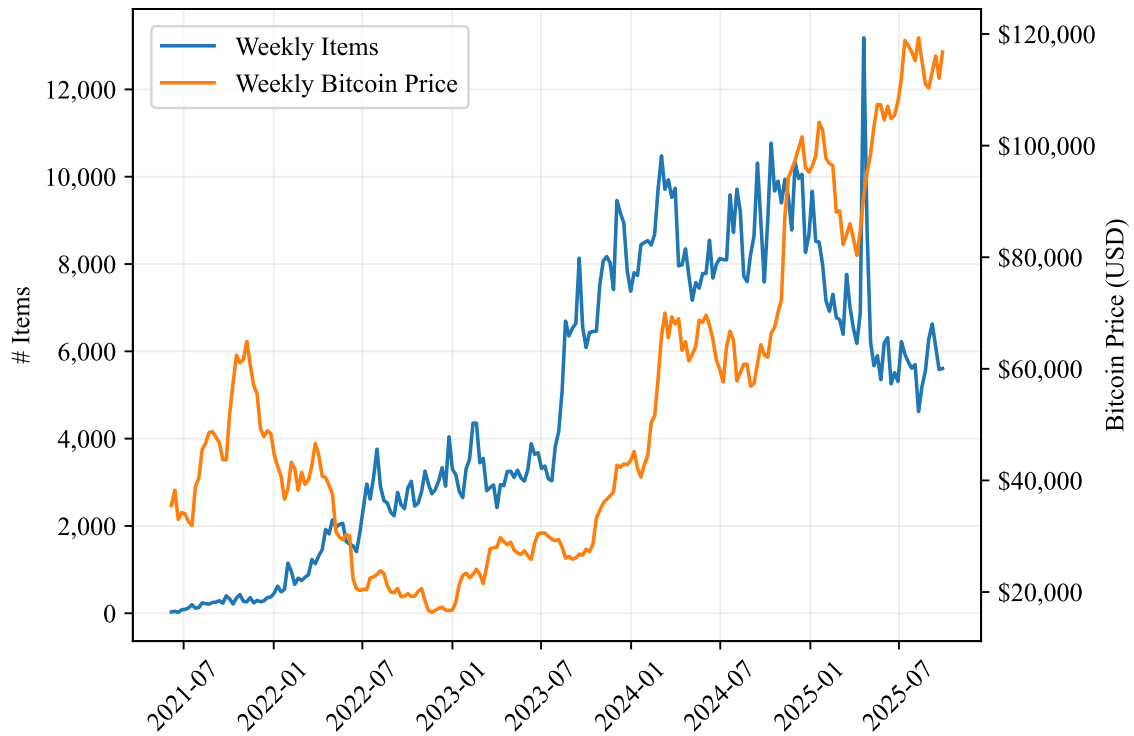
micropayment systems into user-generated content platforms could therefore be an effective approach to managing the quantity-quality tradeoff in content production. Cost-based screening may become especially relevant as platforms seek to discourage AI-generated content submissions. Our results may therefore offer practical guidance to platform designers considering the integration of direct payment features.

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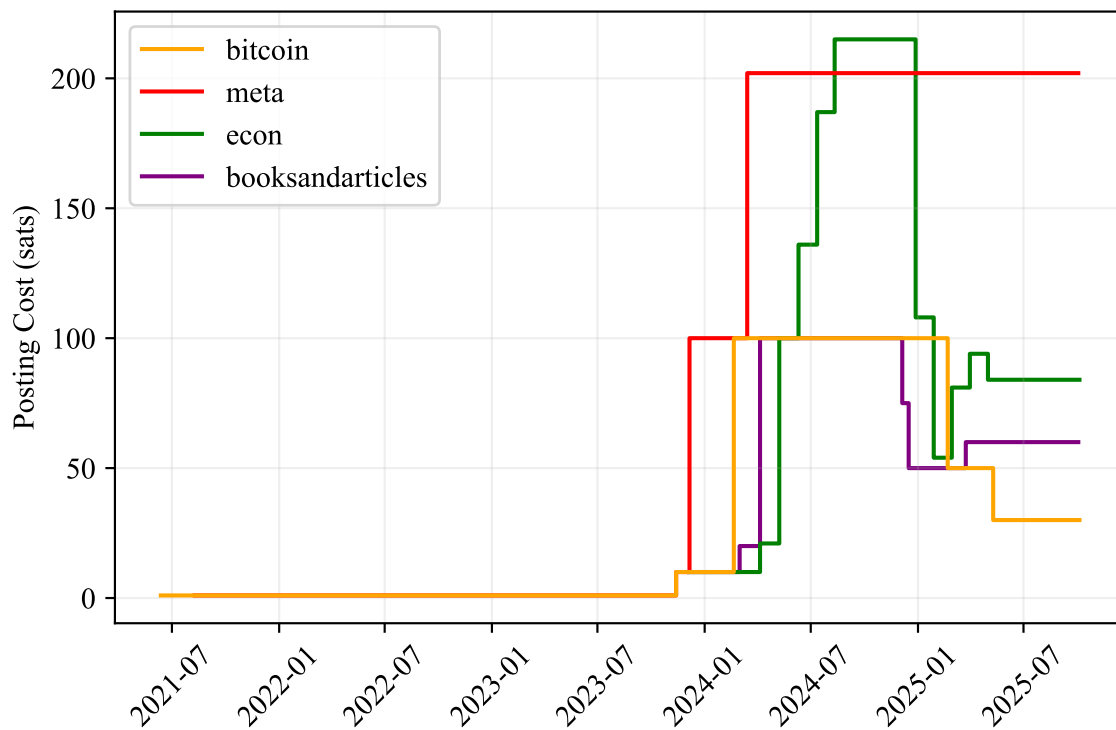
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Figure 1: Weekly Stacker News Items and Bitcoin Price



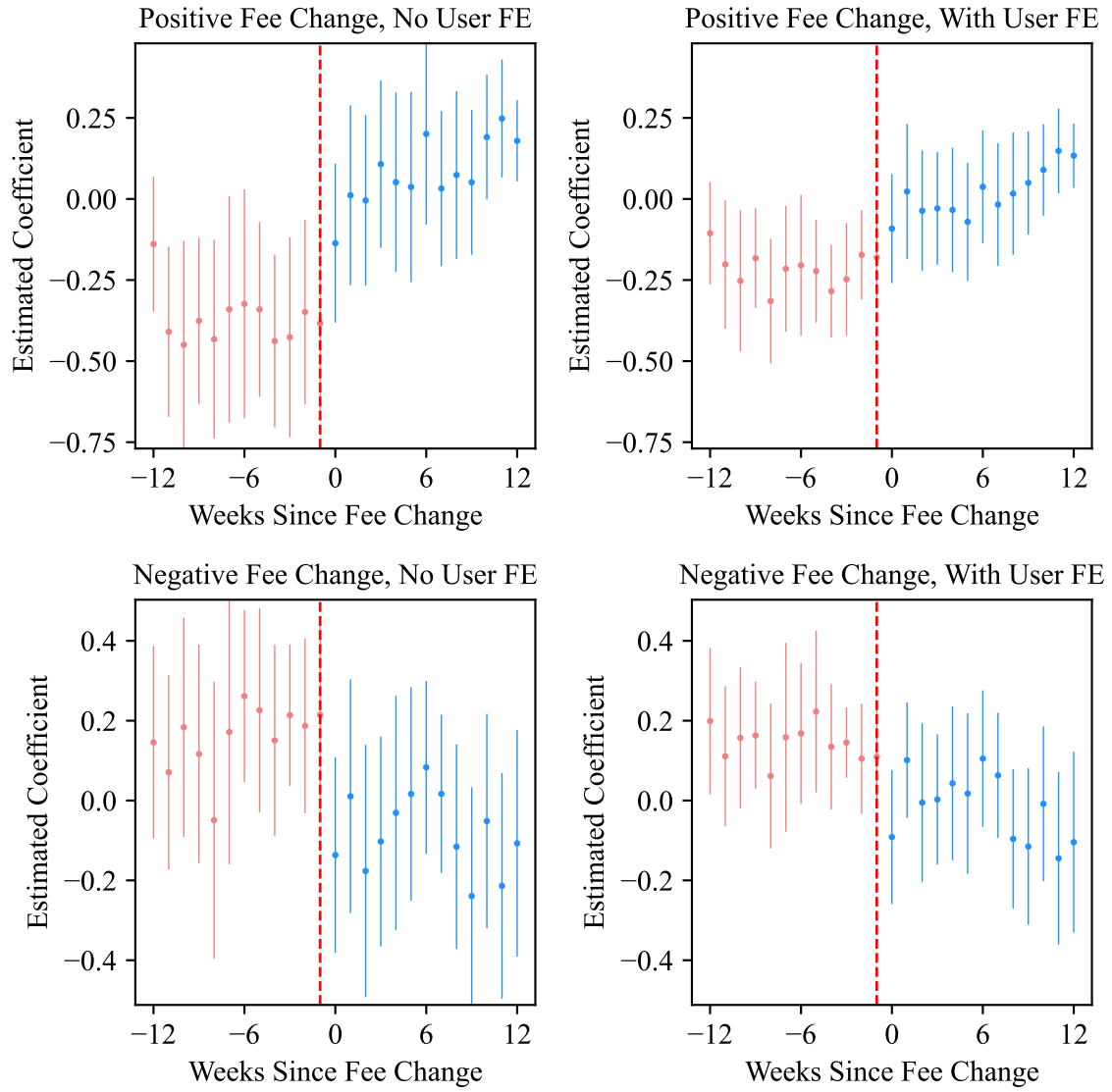
Note: Shows the number of items posted on Stacker News by week along with the weekly Bitcoin price. Weekly Bitcoin price is calculated as the average of the midpoint between daily highs and lows as reported by CoinMarketCap.com.

Figure 2: Posting Cost Histories for Select Territories



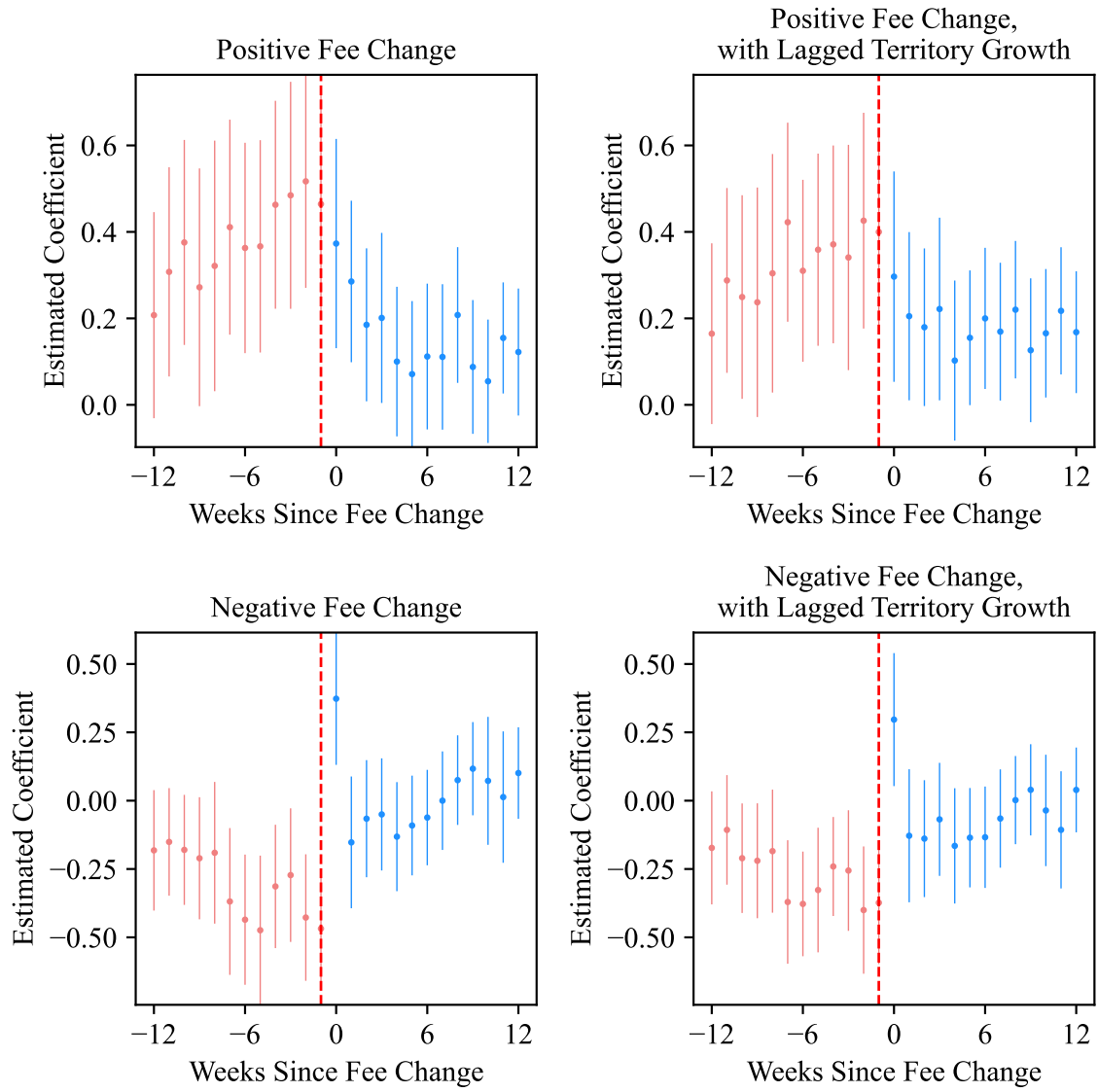
Note: This figure shows the territory fee path for four selected territories. The territories were selected both for their popularity and to highlight the differences in timing and magnitude of fee changes across territories.

Figure 3: Post Quality Event Study



Note: Shows estimated coefficients and 95% confidence intervals from the post quality event study regression specified by equation (5). All regressions shown include territory fixed effects and week fixed effects.

Figure 4: Post Quantity Event Study



Note: Shows estimated coefficients and 95% confidence intervals from the post quantity event study regression specified by equation (7). All regressions shown include territory fixed effects and week fixed effects.

Table 1: Effect of Posting Cost on Post Quality

	<i>Dependent variable:</i> ln(Zaps)			
	(1)	(2)	(3)	(4)
ln(Posting Cost)	0.393*** (0.048)	0.273*** (0.058)	0.208*** (0.045)	0.100** (0.044)
ln(Territory Weekly Posts)			-0.344*** (0.076)	-0.129** (0.063)
Constant	2.530*** (0.076)			
Territory FE	N	Y	Y	Y
Week FE	N	Y	Y	Y
User FE	N	N	N	Y
Observations	179,480	179,477	179,477	178,198
R ²	0.079	0.181	0.185	0.400

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Regression of number of sats a post earns in the first 48 hours on posting cost. The unit of observation is a post. Standard errors are clustered by territory.

Table 2: Effect of Posting Cost on Post Quantity

	<i>Dependent variable:</i> ln(Posts)		
	(1)	(2)	(3)
ln(Posting Cost)	-0.242*** (0.079)	-0.266*** (0.063)	-0.265*** (0.074)
Constant	3.328*** (0.415)		
Territory FE	N	Y	Y
Week FE	N	Y	Y
Territory Operator FE	N	N	Y
Observations	6,023	6,020	6,020
R ²	0.092	0.776	0.788

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Regression of number of posts in a territory on posting costs for that week. The unit of observation is a territory-week. Standard errors are clustered by territory.

Table 3: Effect of Past Quality-Adjusted Zap Signals on Post Count

	<i>Dependent variable:</i> N_{ijw}				
	(1)	(2)	(3)	(4)	(5)
X_{ijw}^U	0.085*** (0.017)			0.273** (0.114)	0.147*** (0.031)
X_{ijw}^R		0.055*** (0.013)		-0.043 (0.029)	
X_{ijw}^E			0.094*** (0.020)	-0.165 (0.113)	
$X_{ijw}^U \times \ln(1 + \sum_{w' < w} N_{ijw'})$					-0.066** (0.026)
User-Territory FE	Y	Y	Y	Y	Y
Territory-Week FE	Y	Y	Y	Y	Y
User-Week FE	Y	Y	Y	Y	Y
Observations	127,854	127,854	127,854	127,854	127,854
Pseudo-R ²	0.706	0.706	0.706	0.706	0.706

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Poisson regression of user i 's post count in territory j in week w on past quality-adjusted zap signals. See Section 5 for details and notation. The unit of observation is a user-territory-week. Standard errors are clustered by territory-week.

Table 4: Effect of Profitability on User Exit

	<i>Dependent variable:</i> Whether user becomes inactive			
	(1)	(2)	(3)	(4)
Unprofitable in last 8 weeks	0.105*** (0.005)	0.062*** (0.005)	0.060*** (0.005)	0.055*** (0.005)
$\dots \times$ BTC price appreciation in last 8 weeks				0.056*** (0.018)
log(Items posted in last 8 weeks)		-0.029*** (0.001)	-0.029*** (0.001)	-0.029*** (0.001)
Constant	0.079*** (0.002)	0.157*** (0.003)		
Week FE	N	N	Y	Y
Observations	92,956	92,956	92,956	92,956
R ²	0.018	0.042	0.048	0.048

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Linear probability model of whether a user becomes inactive in a given week. A user becomes inactive if they start a four or more week consecutive period of no activity. The unit of observation is a user-week. Standard errors are clustered by user.

Online Appendix

A Proofs

Proof of Proposition 1

Proof. A post idea (q, ϵ) is posted if and only if $\epsilon > \kappa - \ln V(q)$. Define $\pi(q, \kappa)$ as the probability (over ϵ) that a post idea of quality q is posted when the log cost is κ :

$$\pi(q, \kappa) = P(\epsilon > \kappa - \ln V(q))$$

$\pi(q, \kappa)$ is the survival function of $\epsilon|q$, evaluated at $\kappa - \ln V(q)$: $\pi(q, \kappa) = S_{\epsilon|q}(\kappa - \ln V(q)|q)$.

(i) The unconditional probability of posting is:

$$P(\text{post}) = \int S_{\epsilon|q}(\kappa - \ln V(q)|q) f_q(q) dq$$

Since $S_{\epsilon|q}(\cdot|q)$ is a survival function, its first derivative is negative everywhere. An increase in c will therefore reduce the value of $S_{\epsilon|q}(\kappa - \ln V(q))$ at every q . Since that is what we are integrating over to get $P(\text{post})$, this strictly reduces $P(\text{post})$. The expected number of posts per unit time is $\gamma P(\text{post})$, so an increase in c strictly reduces the expected number of posts.

(ii) Take any $c' > c$ with corresponding log costs $\kappa' > \kappa$. The set of ideas that were posted at log cost κ can be partitioned into two groups:

- **Surviving posts.** Ideas that are still posted at log cost κ' , i.e. $\ln V(q) + \epsilon > \kappa'$

- **Marginal posts.** Ideas that are posted at κ but not at κ' , i.e. $\kappa < \ln V(q) + \epsilon < \kappa'$.

By the Law of Iterated Expectations,

$$E[q|\text{post at } \kappa] = \alpha E[q|\text{surviving}] + (1 - \alpha)E[q|\text{marginal}]$$

where $\alpha = P(\text{surviving}|\text{post at } \kappa)$. Noting that $E[q|\text{post at } \kappa'] = E[q|\text{surviving}]$, we rearrange the above equation to obtain:

$$E[q|\text{post at } \kappa'] - E[q|\text{post at } \kappa] = (1 - \alpha) (E[q|\text{surviving}] - E[q|\text{marginal}])$$

Thus, to show that expected post quality is increasing in c , it suffices to show that the expected quality of the surviving posts is always higher than the expected quality of the marginal posts.

To demonstrate this, it is sufficient to establish that for any $q_2 > q_1$, the ratio $\pi(q_2, \kappa)/\pi(q_1, \kappa)$ is strictly increasing in κ . Intuitively, this says that as costs rise, higher quality posts retain a larger share of the posting probability relative to lower quality ideas. This would imply that the marginal posts are tilted toward lower quality than the surviving posts.

We write:

$$\frac{\pi(q_2, \kappa)}{\pi(q_1, \kappa)} = \frac{S_{\epsilon|q}(\kappa - \ln V(q_2)|q_2)}{S_{\epsilon|q}(\kappa - \ln V(q_1)|q_1)}$$

Define the hazard rate as $h(\epsilon|q) = f_{\epsilon|q}(\epsilon|q)/S_{\epsilon|q}(\epsilon|q)$. Differentiating $\pi(q_2, \kappa)/\pi(q_1, \kappa)$

with respect to κ shows us that the ratio is increasing in κ if and only if:

$$h(\kappa - \ln V(q_1)|q_1) > h(\kappa - \ln V(q_2)|q_2)$$

The proof now proceeds in two parts.

Claim 1. First, we show that $h(\epsilon|q_1) \geq h(\epsilon|q_2)$ for any $q_2 > q_1$. By the MLRP, for any $t > \epsilon$, we have that $\ell(t; q_1, q_2) \geq \ell(\epsilon; q_1, q_2)$, which rearranges to:

$$f_{\epsilon|q}(t|q_2)f_{\epsilon|q}(\epsilon|q_1) \geq f_{\epsilon|q}(\epsilon|q_2)f_{\epsilon|q}(t|q_1)$$

Integrating both sides over t from ϵ to ∞ gives us:

$$S_{\epsilon|q}(\epsilon|q_2)f_{\epsilon|q}(\epsilon|q_1) \geq S_{\epsilon|q}(\epsilon|q_1)f_{\epsilon|q}(\epsilon|q_2)$$

which rearranges to $h(\epsilon|q_1) \geq h(\epsilon|q_2)$ as was to be demonstrated.

Claim 2. Now, we show that $h(\epsilon|q)$ is increasing in ϵ for each q . Take the derivative of $h(\epsilon|q)$ with respect to ϵ . The sign is equal to the sign of:

$$f_{\epsilon|q}(\epsilon|q)^2 + f'_{\epsilon|q}(\epsilon|q)S_{\epsilon|q}(\epsilon|q)$$

When $f'_{\epsilon|q}(\epsilon|q) \geq 0$, this is immediately positive. When $f'_{\epsilon|q}(\epsilon|q) < 0$, we rely on the log-concavity property. Since $f_{\epsilon|q}(\cdot)$ is log-concave, the following holds for every $t > \epsilon$:

$$\ln f_{\epsilon|q}(t|q) < \ln f_{\epsilon|q}(\epsilon|q) + (t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)}$$

Exponentiating both sides:

$$f_{\epsilon|q}(t|q) < f_{\epsilon|q}(\epsilon|q) \exp \left[(t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)} \right]$$

Now integrating over t from ϵ to ∞ :

$$\begin{aligned} S_{\epsilon|q}(\epsilon|q) &< f_{\epsilon|q}(\epsilon|q) \int_{\epsilon}^{\infty} \exp \left[(t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)} \right] dt \\ &= - \frac{f_{\epsilon|q}(\epsilon|q)^2}{f'_{\epsilon|q}(\epsilon|q)} \end{aligned}$$

Since $f'_{\epsilon|q}(\epsilon|q) < 0$, this can be rearranged to:

$$f_{\epsilon|q}(\epsilon|q)^2 + f'_{\epsilon|q}(\epsilon|q) S_{\epsilon|q}(\epsilon|q) > 0$$

which was to be demonstrated. Thus, $h(\epsilon|q)$ is strictly increasing in ϵ for all q .

By Assumption 3, $\kappa - \ln V(q_1) > \kappa - \ln V(q_2)$. By Claim 2, this implies that $h(\kappa - \ln V(q_1)|q_1) > h(\kappa - \ln V(q_2)|q_1)$. By Claim 1, $h(\kappa - \ln V(q_2)|q_1) \geq h(\kappa - \ln V(q_2)|q_2)$. Together, we have $h(\kappa - \ln V(q_1)|q_1) > h(\kappa - \ln V(q_2)|q_2)$, which was to be demonstrated.

□

B Estimating $\ln v(q)$

B.1 Description of the procedure

Let p index a post and let Z_p be the amount of tips it receives in the first 48 hours.

As described in equation (21), we model Z_p as:

$$\ln Z_p = \mu_{i(p)j(p)} + \underbrace{\delta_{w(p)} + g(X_p)}_{\ln v(q_p)} + \varepsilon_p \quad (21)$$

and we estimate each post's quality adjusted zap signal, $\hat{r}_p$ as:

$$\begin{aligned} \hat{r}_p &= \ln Z_p - \widehat{\ln v(q)} \\ &= \ln Z_p - \hat{\delta}_{w(p)} - \hat{g}(X_p) \end{aligned} \quad (22)$$

This section describes our procedure for estimating equation (21) and thereby recovering estimates for $\hat{\delta}_{w(p)}$ and $\hat{g}(X_p)$. Since it is a mixed model with both linear fixed effects and a non-parametric function of X_p , we estimate it iteratively. Let i now index a turn in the iterative procedure instead of a user. The procedure starts at turn $i = 0$ with $\hat{g}^{(0)}(X_p) = 0$. At each turn $i + 1$, the procedure first uses OLS to estimate the fixed effects:

$$\hat{\mu}_{i(p)j(p)}^{(i+1)}, \hat{\delta}_{w(p)}^{(i+1)} = \text{OLS Regression of } \left(\ln Z_p - \hat{g}^{(i)}(X_p) \right) \text{ on } \mu_{i(p)j(p)}, \delta_{w(p)} \quad (24)$$

and then uses XGBoost to estimate $\hat{g}^{(i+1)}$ using the new fixed effects:

$$\hat{g}^{(i+1)}(X_p) = \text{XGBoost Regression of } \left(\ln Z_p - \hat{\mu}_{i(p)j(p)}^{(i+1)} - \hat{\delta}_{w(p)}^{(i+1)} \right) \text{ on } X_p \quad (25)$$

The procedure can be run a fixed number of times or iterate until convergence. At the end of the procedure, we estimate the fixed effects a final time and use it to compute an estimate for each post’s quality adjusted zap signal, as per equation (22).

B.2 Calculation of embeddings

Text embeddings were computed for each post by inputting the post text into OpenAI’s `text-embedding-3-small` embedding model. The embedding model computes a 1,536 dimensional embedding vector for each post that encodes the meaning of the text in such a way that two posts with similar text will have a small distance between their embedding vectors. The raw embedding dimensions are learned from a large general purpose training corpus and are not tailored to the structure of our dataset. We therefore compute the principal components of the raw embedding vectors using our corpus. The principal components are orthogonal linear combinations of the original 1,536 embedding dimensions that explain the greatest variation across posts in our data. Figure 5 shows the scree plot of the principal components analysis, for the first 30 components. To keep our regression in (25) parsimonious, we decided to use only the first 10 components in the analysis. These components explain 23.7% of the total variation, but our results in Section 5 are robust to the specific choice of how many components to use.

B.3 Estimation results

Table 5 reports the root mean squared error and R^2 statistics for the estimation procedure for three iterations. It also compares the metrics to an OLS benchmark. The non-parametric fixed effects model outperforms the OLS model, and the predictive accuracy converges quickly after a few iterations.

To get a sense of which factors are more important for explaining $\ln Z_p$, we present Table 6. On the left hand side of the table, we show how much each of $\hat{\mu}_{i(p)j(p)}$, $\delta_{w(p)}$, and $\hat{g}(X_p)$ each contribute to the variance of $\widehat{\ln Z_p}$.¹⁷ On the right hand side of the table, we show the “permutation importance” of each of the variables in X_p . The permutation importance measures how much the RMSE would increase if the model was re-estimated using shuffled values for that variable. In computing the permutation importance, we only re-estimate $\hat{g}(X_p)$ at the last iterative step, holding the fixed effect estimates $\hat{\mu}_{i(p)j(p)}$ and $\delta_{w(p)}$ fixed.

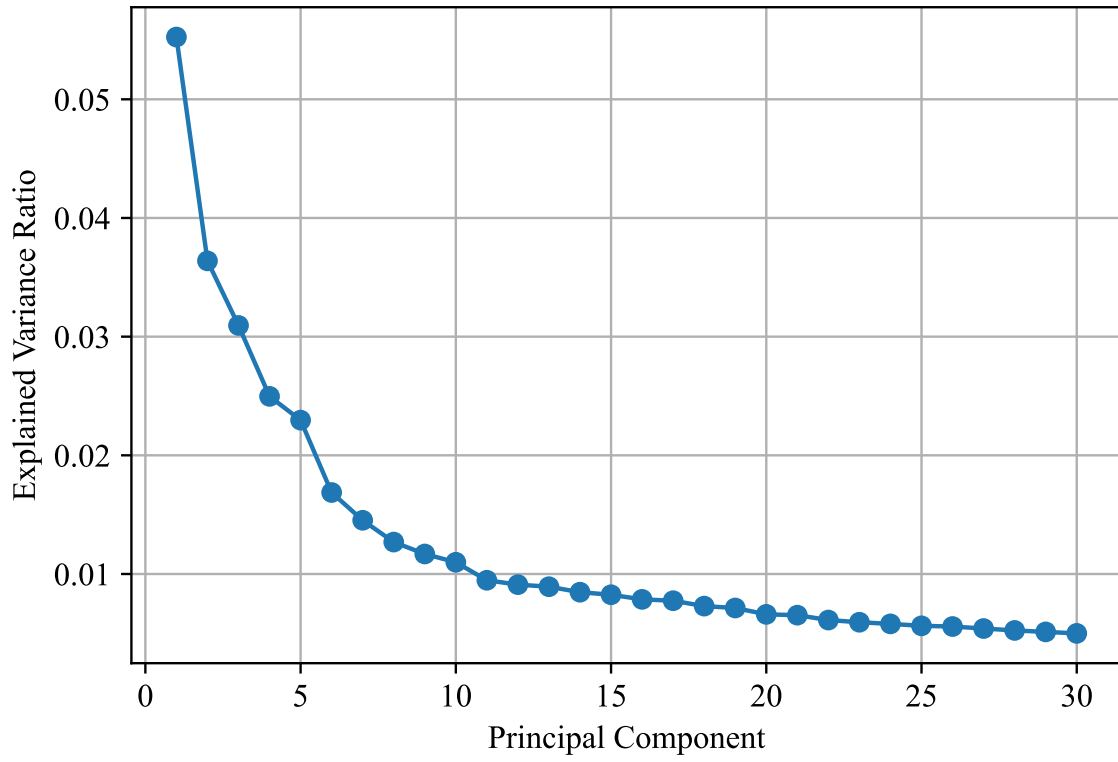
The left half Table 6 shows that user-territory fixed effects explain most of the variance in the predictor, about 76.1%. Week fixed effects explain 13.1% of the variance, and $\hat{g}(X_p)$ explains the remaining 10.7%. Note that the combined model explains only about 50% of the variance in log zaps, so we can halve the previous numbers to estimate how much each components contributes to the variance of log zaps. The relatively small explanatory power of $\hat{g}(X_p)$ suggests that post quality is hard to measure directly from objective attributes like word count, image count, and coarse semantic measures. The relatively high explanatory power of user-territory fixed effects suggests that user-territory match quality is an important factor and that users, to the extent that they are uncertain about their territory-specific match qualities, are incentivized to learn about them.

The right half of Table 6 shows the permutation importance of the variables in X_p . The number of words in the post is the most important factor, followed by the first text embedding dimension, followed by whether or not it is a link post. Instead of trying to interpret the meaning of each embedding dimension, we inspect the posts with highest and lowest values of $\hat{g}(X_p)$ in an attempt to characterize the objective measurements that lead to greater or lesser zaps. We found that the posts in the

¹⁷Note that if $Y = a + b + c$ then the percent of Y ’s variance explained by a is $\text{Cov}(a, Y)/\text{Var}(Y)$.

95th or higher percentiles of $\hat{g}(X_p)$ were mostly long-form ideological or technical posts about Bitcoin or Lightning, posts about Stacker News itself, or posts about monetary economics. Since X_p does not contain territory information, this means that the semantic vectors are able to identify such posts. By contrast, posts in the 5th or lower percentiles of $\hat{g}(X_p)$ were more diverse. They include repetitive daily posts that some users make that others tune out, short posts with barely any text, and posts that look like they might have been generated with AI.

Figure 5: Scree Plot for Post Text Embeddings



Note: Shows the scree plot for post text embeddings, indicating the explained variance ratio for each principal component. The total explained variance at $K = 10$ is 23.7%.

Table 5: Performance of Iterative Fixed Effects XGBoost Model

	RMSE	R ²
First iteration	1.630	0.520
Second iteration	1.618	0.527
Final iteration	1.615	0.529
OLS benchmark	1.681	0.490
Observations	175,857	

Notes: Performance metrics for iterative fixed effects XGBoost model described in Appendix Section B. Regressors include number of words, number of images and links, whether the post is a link post, and 10 dimensional semantic text embeddings. Fixed effects include user-territory effects and week effects.

Table 6: Importance of Factors in Fixed Effects XGBoost Model

Percent of Predictor Variance Explained		Importance of Factor by Permutation Importance	
Estimate	Pct. Explained	Variable	Importance
$\hat{\mu}_{i(p)j(p)}$	0.761	Number of words	0.037
$\hat{\delta}_{t(p)}$	0.131	Text embedding 1	0.023
$\hat{g}(X_p)$	0.107	Link post	0.023
		Text embedding 2	0.021
		Text embedding 5	0.020
		Text embedding 3	0.019
		Text embedding 9	0.018
		Text embedding 7	0.018
		Text embedding 10	0.017
		Text embedding 4	0.017
		Text embedding 8	0.016
		Text embedding 6	0.015
		Number of images/links	0.010

Notes: The left hand side of this table shows the percent of variance of $\widehat{\ln Z_p}$ explained by the corresponding estimate. The right hand side of this table shows the importance of each factor in $\hat{g}(X_p)$ by measuring the RMSE increase that results when the values of that variable are randomly shuffled and the model is re-estimated. See Appendix Section B for details.